USC University of Southern California

Discrete Maths

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Pigeonhole Principle

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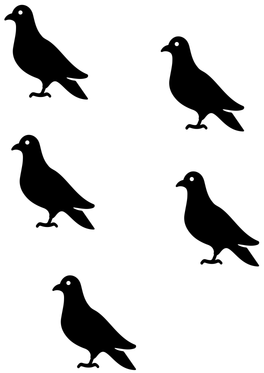
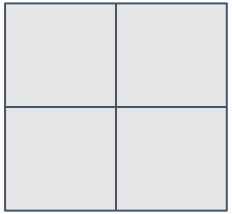
Things $\rightarrow$ Places to put things

Pigeonhole Principle

k pigeons



n holes



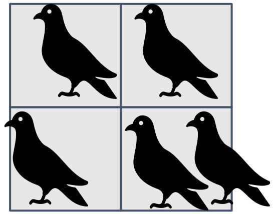
$k > n$

Pigeonhole Principle

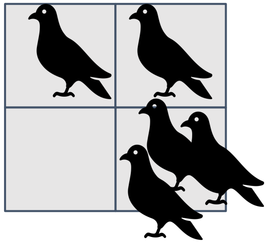
k pigeons

n holes

k > n

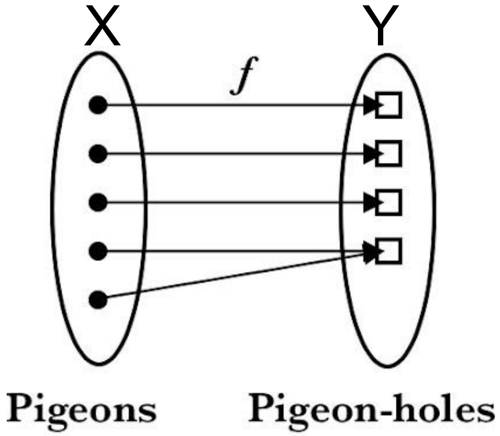


Pigeonhole Principle



Pigeonhole Principle

More formally: if $f : X \to Y$ and $|X| > |Y|$, then there are elements $x_1, x_2 \in X$ such that $x_1 \neq x_2$ and $f(x_1) = f(x_2)$



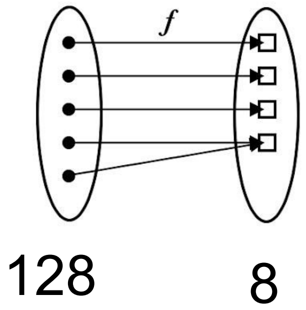
Extended Pigeonhole Principle

For any sets X and Y and any positive integer k such that $|X| > k \cdot |Y|$, if $f : X \to Y$, then there are at least $k + 1$ distinct members $x_1, ..., x_{k+1} \in X$ such that $f(x_1) = ... = f(x_{k+1})$.

Example

- Assume your degree program requires 128 units
- You want to graduate in 8 semesters
- What is the highest number of units you have to take, if you want to minimize the maximum number of semester units

Example



Example

For any sets X and Y and any positive integer k such that $|X| > k \cdot |Y|$, if $f : X \to Y$, then there are at least $k+1$ distinct members $x_1, \ldots, x_{k+1} \in X$ such that $f(x_1) = \ldots = f(x_{k+1})$.

$$|X| > k \cdot |Y|$$

128 8

|X| / |Y| > k

Applications

- Combinatorics
- Random numbers
- Geometry
- etc